LOAN APPROVAL PREDICTION USING MACHINE LEARNING

Project Report

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Submitted As Part of Internship/Project Work

Executive summary

Loan Approval Prediction Using Machine Learning

The objective of this project was to build a machine learning model capable of predicting whether a loan application would be approved or rejected based on applicant information. The Loan Prediction dataset was used for analysis and model development.

The dataset contained 614 records with demographic, financial, and credit-related attributes. Data preprocessing involved handling missing values using median and mode imputation, removing unnecessary columns, and encoding categorical variables into numerical form for machine learning.

Class distribution analysis showed that approximately 68.73% of loan applications were approved and 31.27% were rejected. Since the imbalance was moderate, no resampling techniques such as SMOTE were applied.

Two machine learning models were developed and evaluated: Logistic Regression and Random Forest. Logistic Regression achieved a training accuracy of 79.84%, testing accuracy of 86.18%, and ROC-AUC score of 0.8019. Random Forest achieved a training accuracy of 100% and testing accuracy of 82.93%, indicating overfitting despite strong training performance.

Feature importance analysis identified Credit_History as the most influential factor affecting loan approval decisions, followed by ApplicantIncome, LoanAmount, and CoapplicantIncome.

Based on model evaluation metrics and generalization performance, Logistic Regression was selected as the recommended model for deployment. The model provides reliable predictions and can assist financial institutions in streamlining loan approval decisions.

Results and findings

Dataset Size: 614 records

Target Variable:

- Loan_Status

Class Distribution:

- Approved: 68.73%
- Rejected: 31.27%

Logistic Regression:

- Training Accuracy: 79.84%
- Testing Accuracy: 86.18%
- ROC-AUC Score: 0.8019

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Confusion Matrix:
[[22 16]
 [ 1 84]]

Classification Report:
              precision    recall  f1-score   support

           0       0.96      0.58      0.72        38
           1       0.84      0.99      0.91        85

    accuracy                           0.86       123
   macro avg       0.90      0.78      0.81       123
weighted avg       0.88      0.86      0.85       123
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ROC-AUC Score: 0.8018575851393189
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Random Forest:

- Training Accuracy: 100%
- Testing Accuracy: 82.93%
- ROC-AUC Score: 0.7950

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Confusion Matrix:
[[24 14]
 [ 7 78]]

Classification Report:
              precision    recall  f1-score   support

           0       0.77      0.63      0.70        38
           1       0.85      0.92      0.88        85

    accuracy                           0.83       123
   macro avg       0.81      0.77      0.79       123
weighted avg       0.83      0.83      0.82       123

ROC-AUC Score: 0.795046439628483
```

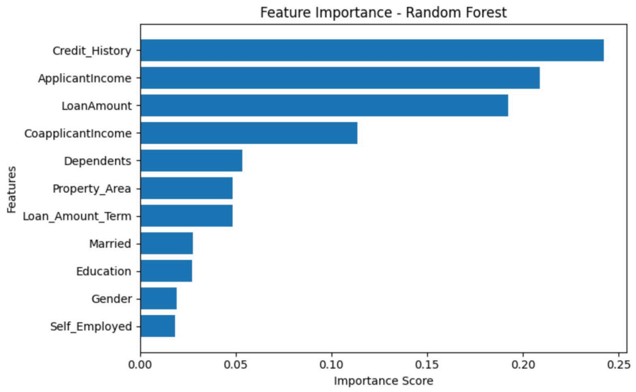
Key Findings:

- Logistic Regression achieved the highest testing accuracy.

- Random Forest showed signs of overfitting.

Feature Importance Analysis:

- Credit_History was the most important feature influencing loan approval decisions.
- ApplicantIncome, LoanAmount, and CoapplicantIncome also contributed significantly.



Final Recommendation:

Deploy the Logistic Regression model with a decision threshold of 0.5 for loan approval prediction.

Conclusion & Recommendation

Conclusion

A Loan Approval Prediction model was successfully developed using machine learning techniques. After preprocessing the data and evaluating multiple models, Logistic Regression emerged as the best-performing model with a testing accuracy of 86.18% and a ROC-AUC score of 0.8019.

Random Forest achieved perfect training accuracy but showed signs of overfitting, resulting in lower generalization performance on the test data.

Feature importance analysis revealed that Credit_History is the most influential factor affecting loan approval decisions, followed by ApplicantIncome, LoanAmount, and CoapplicantIncome.

Recommendation

Based on the evaluation results, Logistic Regression is recommended for deployment due to its superior predictive performance and better generalization capability. A decision threshold of 0.5 is suggested for loan approval prediction.